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# CSE 250 Summer 2026

## Assignment 02

Section 10A [Tarvin Anjum Aditto]

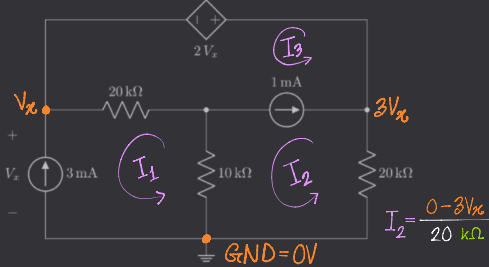
Circuits & Electronics

30<sup>th</sup> July 2026  
10:30 p.m.

Deadline: 31<sup>st</sup> July 2026  
3:29 p.m.

### Problems for assignment 2:

1. Using mesh analysis to find the power of the dependent voltage source. Do you think nodal analysis would have been easier? If yes, why so? [5 marks]



$$I_3 - I_2 = 1 \text{ mA}$$

$$I_1 = -3 \text{ mA}$$

writing out the KCL equations are much easy here

KVL @ supermesh  $m_2 - m_3$ :

$$10(I_2 - I_1) + 20I_2 + 2V_x + 20(I_3 - I_1) = 0$$

→ mesh analysis is harder, because, we have to find a way to translate  $V_x$  into  $I_1/2/3$

$$\begin{aligned} 10(I_2 + 3) + 20I_2 + 2V_x + 20(I_3 + 3) &= 0 \\ 30 + 30I_2 + 2 \cdot \frac{-20}{3} I_2 + 20I_3 + 60 &= 0 \\ 90 + 90I_2 - 40I_2 + 60I_3 + 180 &= 0 \\ 50I_2 + 60I_3 &= -270 \end{aligned}$$

$$I_2 = \frac{0 - 3V_x}{20 \text{ k}\Omega}$$

$$V_x = \frac{-20}{3} I_2$$

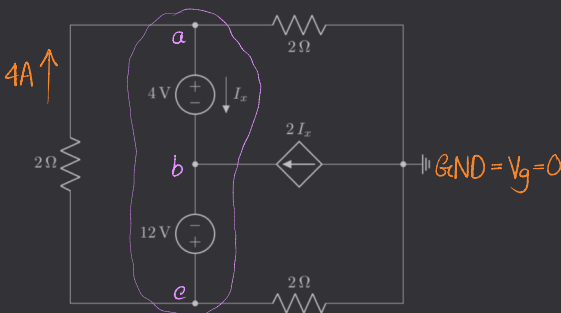
Solving Simultaneously:

$$I_3 = -2 \text{ mA} \quad I_2 = -3 \text{ mA} \quad I_1 = -3 \text{ mA}$$

$$V_x = 20$$

∴ Power of the dependant Voltage Source  
=  $+I_3 \cdot 2V_x$   
=  $-80 \text{ milliWatt?}$  ∴ Supplying.

2. Find  $I_x$  in the circuit shown above using Nodal analysis. Do you think mesh analysis would have been easier? If yes, why so? [3 marks]  
(b) Which side (top or bottom) of the  $2\Omega$  resistor on the left has a higher voltage? [1 mark]  
(c) Determine the voltage across the  $2I_x$  dependent current source. [1 mark]



$$V_a - V_b = 4 \quad (i)$$

$$V_c - V_b = 12 \quad (ii)$$

KCL @ Ground Node assume outgoing cur

$$2I_x + \frac{V_b - V_a}{2} + \frac{V_b - V_c}{2} = 0$$

$$I_x = \frac{0 - V_a}{2} + \frac{V_c - V_a}{2}$$

$$4I_x - V_a - V_c = 0$$

$$2I_x = -V_a + V_c - V_a$$

$$2V_c - 4V_a - V_a - V_c = 0$$

$$2I_x = V_c - 2V_a$$

$$V_c - 5V_a = 0 \quad (iii)$$

KCL @ a-b-c supernode: — gets us the same eq. as (iii)

$$\frac{V_a - 0}{2} + \frac{V_c - 0}{2} = 2I_x$$

$$V_a + V_c = 4I_x = 2V_c - 4V_a$$

$$5V_a - V_c = 0$$

Solving Simultaneously:

$$V_a = 2 \quad V_b = -2 \quad V_c = 10$$

$$2I_x = 10 - 4 = 6$$

$$I_x = 3$$

b)  $V_c$  = bottom side has higher voltage

c) Voltage Across  $2I_x$  dependent current source is 2Volt  
and current is flowing from high → low.

2 unknown mesh currents.

∴ mesh analysis would have been same hardness.

∴  $2I_x$  is absorbing.